THE UPPER SHRIEK OF A DUALIZING COMPLEX IS DUALIZING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AA3 (sources/stacks-duality/pinned-original-source.tex line 4708).

Allowed standard prerequisites (editorial).

- The shriek of the affine line and dualizingness for polynomial rings. Same chapter, Lemma labelled lemma-shriek-affine-line, cited at proof lines 4740-4742, and Dualizing Complexes, Lemma labelled dualizing-lemma-dualizing-polynomial-ring, cited at proof lines 4742-4744. Handles the projections.
- The shriek of a closed immersion and dualizingness for quotients. Dualizing Complexes, Lemma labelled dualizing-lemma-dualizing-quotient, cited at proof lines 4746-4748, with Same chapter, Lemma labelled lemma-shriek-closed-immersion and Lemma labelled lemma-sheaf-with-exact-support-quasi-coherent, cited at proof lines 4744-4748. Handles the closed immersion.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA1: sources/stacks-duality/pinned-original-source.tex, source lines 4609–4609. Same-chapter pointer into the author environment lemma at source line 4609; invoked at proof line 4721 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A7J: sources/stacks-chow/tags.txt, source lines 12882–12882. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4722 of the counted proof. It is

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retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A7I: sources/stacks-chow/tags.txt, source lines 12881–12881. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 4726 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A78: sources/stacks-duality/pinned-original-source.tex, source lines 2373–2373. Same-chapter pointer into the author environment lemma at source line 2373; invoked at proof line 4728 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA2: sources/stacks-duality/pinned-original-source.tex, source lines 4624–4624. Same-chapter pointer into the author environment lemma at source line 4624; invoked at proof line 4729 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. If K is a dualizing complex for Y, then $f^!K$ is a dualizing complex for X.*

Proof. The question is local on X hence we may assume that X and Y are affine schemes. In this case we can factor $f : X \to Y$ as

$$X \xrightarrow{i} \mathbf{A}^n_Y \to \mathbf{A}^{n-1}_Y \to \ldots \to \mathbf{A}^1_Y \to Y$$

where i is a closed immersion. By Lemma 0AA1 and Dualizing Complexes, Lemma 0A7J and induction we see that the $p^!K$ is a dualizing complex on $\mathbf{A}^n_Y$ where $p : \mathbf{A}^n_Y \to Y$ is the projection. Similarly, by Dualizing Complexes, Lemma 0A7I and Lemmas 0A78 and 0AA2 we see that $i^!$ transforms dualizing complexes into dualizing complexes. □